Polylogarithmic Natural-Density Descent for the Collatz Map

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Version: 3.1 research draft, moving-endpoint first-passage refinement

Content draft: August 2026

2020 Mathematics Subject Classification: Primary 11B83; Secondary 37P99, 60G40

Keywords: Collatz map, natural density, first passage, stopping map, parity vectors, polylogarithmic descent

Abstract

Let

$$T(n) = \begin{cases} n/2, & n \equiv 0 \pmod 2, \\ (3n+1)/2, & n \equiv 1 \pmod 2, \end{cases} \qquad T_{\min}(n) = \min_{k \ge 0} T^k(n).$$

Put

$$\varkappa_* = 1 - H_2(\log_3 2), \qquad A_{FP} = \frac{1}{2(1 - H_2(\log_3 2))} = 9.9911133419\ldots, \qquad c_* = \frac{2}{\log(4/3)} = 6.9521189935\ldots,$$

where H_2 is binary entropy. For every fixed $D > 0$, $c > c_*$, and $\beta > 0$, and every $0 < \gamma < D\varkappa_*$, all but $O_{D,c,\beta,\gamma}(X/(\log\log\log X)^{\gamma})$ integers $n \le X$ possess a shortcut-Collatz iterate, before $c \log n$ steps, satisfying

$$T^k(n) \le C_{tar}(\log n)^{A_{FP}}(\log\log n)^{2A_{FP}}(\log\log\log n)^D, \qquad \max_{0 \le j \le k} T^j(n) \le n^{1+\beta}.$$

More generally, the last factor may be replaced, on a natural-density-one set, by any prescribed function of $\log\log n$ tending to infinity. For every fixed $A > A_{FP}$, the simpler target $C_{tar}(\log n)^A$ has exceptional count $O_{A,c,\beta,\gamma}(X/(\log X)^{\gamma})$ for every $0 < \gamma < \varkappa_*(A - A_{FP})$. The pure target $C(\log n)^{A_{FP}}$, and the displayed critical secondary scale with a bounded final multiplier, are not asserted.

For the weaker target $\exp((\log n)^{1-\delta})$, the same method gives the sharper exceptional count

$$O_{\delta,c,\beta}(X \exp(-\gamma_{\delta,c,\beta}(\log X)^{1-\delta}))$$

for every fixed $0 < \delta < 1$, $c > c_*$, and $\beta > 0$; the same witnesses give every clock constant greater than $3/\log(4/3)$ for the unaccelerated Collatz map.

The proof uses exact parity-cube counting only to certify an all-prefix orbit barrier. It then proceeds deterministically. Strictly nested thresholds turn every later certification failure into a direct first passage from the original shell, while shrinking high-rank barriers confine its feasible cumulative times to $O(\sqrt{M \log M})$ values. Transport over this smaller time support replaces a linear horizon loss by a square-root loss and yields the displayed exponent. The result is an almost-all statement and makes no claim for every starting value.

1. Introduction and main results

We ask how small a value can be forced along the Collatz orbit of almost every initial integer in ordinary natural density, while retaining a logarithmic witnessing time and a bound on the orbit before that witness.

We work throughout with the shortcut Collatz map T displayed in the abstract. All unqualified logarithms are natural, $\log_2$ denotes the base-two logarithm, and $\mathbb{N} = \{1, 2, 3, \dots\}$. Put

$$a_0 = \frac{\log_2 3}{2}. \tag{1.1}$$

Thus $a_0 - 1$ is the mean base-two logarithm of the multiplicative main term under the exact uniform parity coding used below; the affine correction is treated separately in Section 3.

Define binary entropy by

$$H_2(p) = -p\log_2 p - (1-p)\log_2(1-p),$$

and put

$$p_* = \log_3 2, \qquad A_{FP} = \frac{1}{2(1-H_2(p_*))}, \qquad c_* = \frac{2}{\log(4/3)}. \tag{1.2}$$

Where the constants come from

The two constants in (1.2) record the two quantitative costs of the proof. First,

$$1 - a_0 = \frac{\log_2(4/3)}{2}, \qquad c_* = \frac{1}{(1-a_0)\log 2}.$$

Thus $1 - a_0$ is the mean base-two logarithmic loss of the multiplicative main term per shortcut step, and $c_* \log n$ is the corresponding mean-drift clock for clearing $\log_2 n$ bits. This interpretation does not assert a matching lower bound or optimality theorem. Second, put

$$\varkappa_* = 1 - H_2(p_*).$$

The entropy rates available from Proposition 3.3 approach $\varkappa_* \log 2$. The sharp terminal binomial prefactor and the compressed feasible-time factor from Lemma 6.1 leave the rank budget

$$\varkappa_* L_M - \frac{1}{2}\log_2 M - \log_2 \log M.$$

Its divergence is the exact endpoint condition used below. For $L_M \sim A\log_2 M$, the leading power changes sign at $A\varkappa_* = 1/2$, giving $A_{FP} = 1/(2\varkappa_*)$. At equality the remaining logarithmic budget forces the secondary factor displayed in Corollary 1.2. Thus A_{FP} is a threshold of the present argument rather than an intrinsic constant of the Collatz map.

Theorem 1.1 (moving polylogarithmic endpoint)

Let $(A_M)_{M\geq 0}$ be a bounded real sequence and put

$$L_M = \lceil A_M \log_2(M+2) \rceil, \qquad \Delta_M = \varkappa_* L_M - \frac{1}{2}\log_2(M+2) - \log_2 \log(M+3). \tag{1.3}$$

Assume

$$\Delta_M \longrightarrow +\infty . \tag{1.4}$$

For every fixed $c > c_*$ and $\beta > 0$, there are constants $C_{tar}, C_{exc}, \varepsilon > 0$ and M_0 such that the following holds. Let E_M be the set of $n \in I_M$ for which no integer $0 \leq k < c \log n$ simultaneously satisfies

$$T^k(n) \leq C_{tar}(\log n)^{A_M}, \qquad \max_{0\leq j\leq k} T^j(n) \leq n^{1+\beta}.$$

Then, for every $M \geq M_0$,

$$\boxed{\frac{\# E_M}{2^M} \leq C_{exc}\left(2^{-\Delta_M} + M^{-\varepsilon}\right).} \tag{1.5}$$

Consequently these simultaneous witnesses exist on a set of natural density one.

The ceiling in L_M changes Δ_M by only $O(1)$. Hence, when $A_M \to A_{FP}$, condition (1.4) is equivalently

$$\varkappa_*(A_M - A_{FP}) \log M - \log\log M \longrightarrow +\infty . \tag{1.6}$$

The theorem is stated through a moving exponent because this single condition contains both every fixed exponent above A_{FP} and the exact critical secondary scale.

Corollary 1.2 (explicit endpoint profiles)

Fix $c > c_*$ and $\beta > 0$. In each of the following statements the landing has a constant $C_{tar} > 0$, occurs before $c \log n$ shortcut steps, and its witness satisfies $\max_{j\leq k} T^j(n) \leq n^{1+\beta}$.

1. For every fixed $A > A_{FP}$ and every $0 < \gamma < \varkappa_*(A - A_{FP})$, the target

   $$C_{tar}(\log n)^A$$

   fails for at most $O_{A,c,\beta,\gamma}(X/(\log X)^\gamma)$ integers $n \leq X$.

2. For every fixed $B > 1/\varkappa_* = 2A_{FP}$ and every $0 < \gamma < B\varkappa_* - 1$, the target

   $$C_{tar}(\log n)^{A_{FP}}(\log\log n)^B$$

   fails for at most $O_{B,c,\beta,\gamma}(X/(\log\log X)^\gamma)$ integers $n \leq X$.

3. For every fixed $D > 0$ and every $0 < \gamma < D\varkappa_*$, the target

   $$C_{tar}(\log n)^{A_{FP}}(\log\log n)^{2A_{FP}}(\log\log\log n)^D \tag{1.7}$$

   fails for at most $O_{D,c,\beta,\gamma}(X/(\log\log\log X)^\gamma)$ integers $n \leq X$.

More generally, the final factor in (1.7) may be replaced, on a natural-density-one set, by $\Omega(\log\log n)$ for any prescribed function $\Omega(x) \to \infty$. The constants and the retained set may depend on Ω, c, β.

Every target above is $o(n)$, so it is a genuine descent for all sufficiently large n. The assertions are density-one rather than pointwise. Neither the landing constant C_{tar} nor the onset of the asymptotic comparison is made effective.

Theorem 1.3 (endpoint-rate stretched-logarithmic descent)

For every fixed $0 < \delta < 1$, $c > c_*$, and $\beta > 0$, there are constants $C_{\delta,c,\beta}, \gamma_{\delta,c,\beta}, X_{\delta,c,\beta} > 0$ with the following property. For every $X \geq X_{\delta,c,\beta}$, let $E_{\delta,c,\beta}(X)$ be the set of integers $1 \leq n \leq X$ for which no integer $0 \leq k < c \log n$ simultaneously satisfies

$$T^k(n) \leq \exp((\log n)^{1-\delta}), \qquad \max_{0 \leq j \leq k} T^j(n) \leq n^{1+\beta}.$$

Then

$$\# E_{\delta,c,\beta}(X) \leq C_{\delta,c,\beta} X \exp(-\gamma_{\delta,c,\beta}(\log X)^{1-\delta}). \tag{1.8}$$

The endpoint $\delta = 1$ is not asserted.

Corollary 1.4 (raw clock)

Put

$$\mathrm{Col}(n) = \begin{cases} n/2, & n \equiv 0 \pmod 2, \\ 3n+1, & n \equiv 1 \pmod 2. \end{cases}$$

For every fixed $\beta > 0$ and

$$c_{\mathrm{raw}} > \frac{3}{\log(4/3)}, \tag{1.9}$$

the conclusions and exceptional-set estimates in Theorem 1.1, Corollary 1.2, and Theorem 1.3 remain valid for Col, with an integer $\ell < c_{\mathrm{raw}} \log n$ and the raw-orbit ceiling

$$\max_{0 \leq j \leq \ell} \mathrm{Col}^j(n) \leq n^{1+\beta}.$$

Since $3/\log(4/3) = 10.42817849\ldots$, the explicit constant 10.44 remains admissible.

Corollary 1.5 (fixed powers and the graded clock)

For every fixed $\alpha > 0$, the exceptional set for $T_{\min}(n) > n^{\alpha}$ is

$$O_{\alpha,\sigma}(X \exp(-c_{\alpha,\sigma}(\log X)^{\sigma})) \tag{1.10}$$

for every fixed $0 < \sigma < 1$.

If $0 < \alpha < 1$, then for every fixed $\varepsilon > 0$ the same fixed-power target is also reached on a natural-density-one set before

$$\left(\frac{2(1-\alpha)}{\log(4/3)} + \varepsilon\right) \log n \tag{1.11}$$

shortcut steps. Thus the clock decreases continuously with the amount of logarithmic height that remains: for example, the limiting coefficient at $\alpha = 1/2$ is $1/\log(4/3) = 3.47605949\ldots$, rather than the full descent coefficient c_*.

Proof architecture

The proof has two sharply separated parts and six load-bearing steps. Its probabilistic input is confined to exact parity-cube counting; from first-passage reversal onward, the transport and re-certification argument is deterministic.

Exact probabilistic certification.

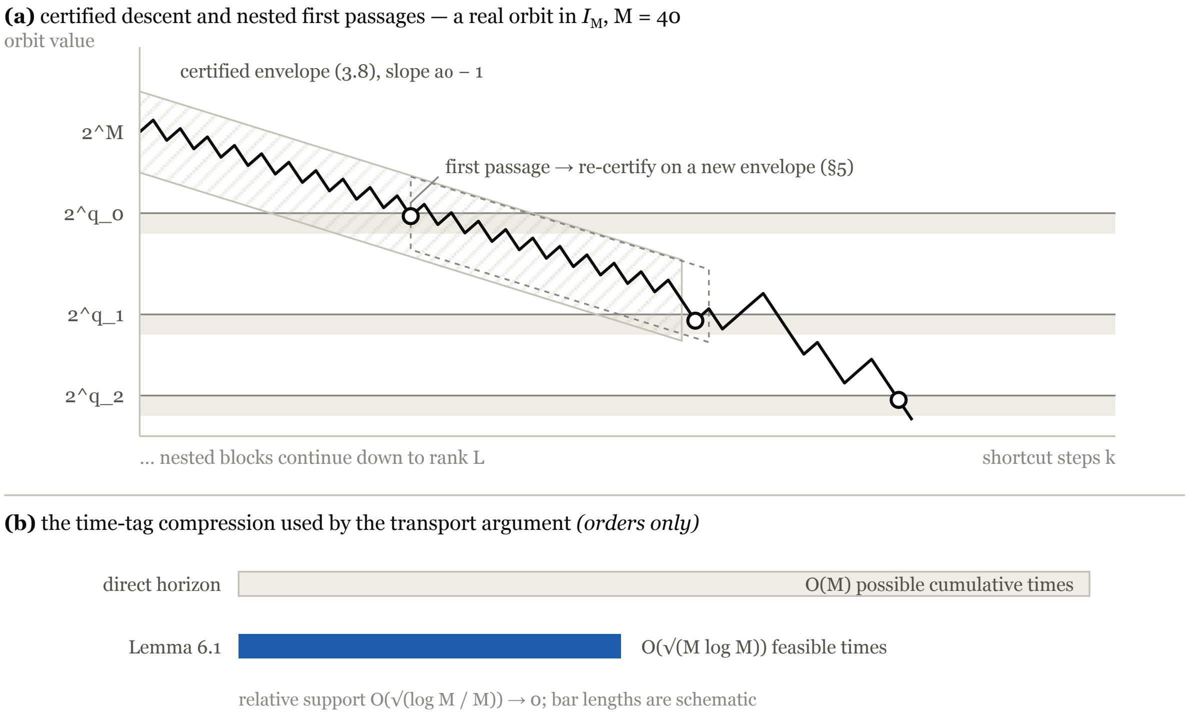
1. The parity-vector bijection turns a complete dyadic shell into the uniform Boolean cube.

2. An adjustable maximal Boolean barrier gives an entropy-sharp shell exceptional rate and a certified all-prefix orbit envelope.

Deterministic Collatz transport and assembly.

3. First-passage reversal gives a loss-filtered tagged-fiber bound for every target subset of a landing shell.
4. Strictly decreasing threshold ranks identify each recursively generated landing with a direct first passage from the original source shell.
5. Rank-dependent high barriers confine every cumulative first-bad time to a common support of size $O(\sqrt{M \log M})$.
6. Support-sensitive transport and a moving low-rank barrier give the exact rank buffer Δ_M, whose positive divergence yields the critical principal exponent A_{FP}.

Two terminal profiles serve different conclusions. The one-regime profile in Theorem 5.3 yields the endpoint-rate stretched-logarithmic companion, whereas the compressed-support profile in Theorem 6.3 yields the strict fixed-exponent profile, and Proposition 6.4 sharpens it to the moving endpoint used in the headline.



The proof architecture. Panel (a) follows an actual shortcut-Collatz orbit inside one certified all-prefix envelope and marks the nested first-passage landings at which the deterministic argument re-certifies. Panel (b) records the proved order reduction in the number of feasible cumulative time tags; bar lengths are schematic and do not encode the unspecified constant in Lemma 6.1.

Relation to previous almost-all results

The preceding endpoint-transport preprint [7] proves the range $0 < \delta < 0.251245530155874\ldots$ by a fixed-time endpoint-fiber/Rényi estimate followed by endpoint iteration. The present proof is independent of that theorem. It replaces fixed-time endpoint transport by all-prefix certification, first-passage reversal, nested direct re-certification, and support-sensitive time aggregation.

The main obstruction is not the contraction of one typical block, but the transport of sparse certification failures through a generated sequence of landings. Direct aggregation over the available time tags incurs a linear horizon loss. Step 5 above compresses those tags to $O(\sqrt{M \log M})$ possibilities; this linear-to-square-root reduction is the quantitative pivot of the paper.

The published results of Korec and Tao, the preprint of Inselmann, and the two recent Tao-to-natural-density bridge manuscripts are compared below only on the coordinates used in their stated theorems. Publication status and proof lineage are recorded in the surrounding prose and references rather than in the table.

Work	Density	Target reached for almost all starts	Clock or quantitative feature
Korec [3]	natural	n^{θ}, every fixed $\theta > a_0$	no clock or quantitative exception used here
Inselmann [2]	natural	n^{ε}, every fixed $\varepsilon > 0$	$2\log n/\log(4/3)$ shortcut steps; density convergence
Tao [5]	logarithmic	every $f(n) \to \infty$	no single global clock in the headline theorem; logarithmic-density estimate
Mazur [4]	natural, bridged from Tao's logarithmic-density framework	every $f(n) \to \infty$	$< 436\log n$ unaccelerated steps; fixed-target logarithmic rate
Allikvere [1]	natural, bridged from Tao's logarithmic-density framework	every $f(n) \to \infty$	$< 12\log n$ unaccelerated steps; $O((\log N_0)^{-1/29} + X^{-1/2000})$ for a fixed target
This paper	natural	$(\log n)^{A_{FP}}(\log\log n)^{2A_{FP}}\Omega(\log\log n)$, every $\Omega \to \infty$; also every fixed $(\log n)^{A}$, $A > A_{FP}$	every shortcut constant $> c_*$; every unaccelerated constant $> 3/\log(4/3)$; target-dependent quantitative rates and a same-witness ceiling

Theorem 1.1 additionally retains the same-witness ceiling $\max_{j \le k} T^{j}(n) \le n^{1+\beta}$. That ceiling is not part of the headline statements in [1] or [4]; this is a comparison of stated outputs, not a claim that their proof-internal trajectory estimates cannot yield related bounds.

The moving polylogarithmic target is smaller than every fixed power and every fixed stretched-logarithmic target in Theorem 1.3, but it still contains the fixed divergent core $(\log n)^{A_{FP}}(\log\log n)^{2A_{FP}}$. It is therefore weaker than an arbitrary diverging function. At the target-only level, Tao's theorem specializes to every target displayed here, albeit in logarithmic rather than natural density. Thus the present theorem does not supersede the arbitrary-threshold conclusions of [1] or [4]. Conversely, those target statements do not supply the present natural-density rates, logarithmic clock, or same-witness ceiling, and the clock scale already appears in [2]. These axes are deliberately not collapsed into a single ordering.

Allikvere conditions Tao's Syracuse/Fourier inputs on the total valuation and supplies a uniform-measure transfer. Mazur starts from the same logarithmic-density frontier and adds a quantitative phase-gap and two-adic lift. Their shared bridge architecture differs from the exact Boolean-cube certification and deterministic first-passage transport separation displayed above.

2. Density, parity, and affine iterates

The argument uses three common coordinates: a shell-to-prefix summation rule, exact parity coding on each dyadic shell, and an affine formula for the iterate. This section establishes them for the certification and transport arguments in Section 3 and Section 4.

For $M \ge 0$, write

$$I_M = [2^{M}, 2^{M+1}) \cap \mathbb{N}$$

for the M-th dyadic shell.

For $S \subseteq \mathbb{N}$, put

$$B_S(X) = \#\{1 \le n \le X : n \notin S\}.$$

We call S (C, D)**-dense** if

$$B_S(X) \le CX^{1-D} \qquad (X \ge 1), \tag{2.1}$$

where $C > 0$ and $0 < D \le 1$. Every such set has natural density one.

We use the following varying-rate shell summation.

Lemma 2.1 (varying-rate dyadic summation)

Let $0 < \sigma \le 1$, $A, \gamma > 0$, and suppose that, for all sufficiently large M,

$$\#(E \cap [2^M, 2^{M+1})) \le Ae^{-\gamma(M+4)^\sigma} 2^M. \tag{2.2}$$

Then there are $\gamma' > 0$ and X_0 such that

$$\#(E \cap [1, X]) \le (2A + 1)Xe^{-\gamma'(\log X)^\sigma} \qquad (X \ge X_0). \tag{2.3}$$

More generally, if a nonnegative sequence $e_M \to 0$ satisfies

$$\#(E \cap I_M) \le e_M 2^M \tag{2.3a}$$

for all sufficiently large M, then E has natural density zero.

Proof

Let $2^J \le X < 2^{J+1}$. The shells below $J/2$ contain fewer than $X^{1/2}$ integers. On the remaining shells,

$$M + 4 \ge \frac{\log X}{4 \log 2}.$$

Thus their total contribution is at most

$$2AX \exp\left(-\frac{\gamma}{(4\log 2)^\sigma}(\log X)^\sigma\right).$$

Choose

$$\gamma' = \min\left\{\frac{\gamma}{(4\log 2)^\sigma}, \frac{1}{2}\right\}.$$

For all sufficiently large X, one has $X^{1/2} \le Xe^{-\gamma'(\log X)^\sigma}$. Adding the early and late contributions proves (2.3).

For the final assertion, fix $\epsilon > 0$ and choose M_0 so that $e_M \le \epsilon$ for $M \ge M_0$. The finitely many earlier shells contribute $o(X)$. If $2^J \le X < 2^{J+1}$, the remaining shells contribute at most

$$\epsilon \sum_{M=M_0}^{J} 2^M \le 2\epsilon X.$$

Letting $\epsilon \downarrow 0$ proves natural density zero. □

For $n \ge 1$, define

$$p_i(n) = 1_{\{T^i(n)\ \mathrm{odd}\}}, \qquad s_k(n) = \sum_{i=0}^{k-1} p_i(n). \tag{2.4}$$

The exact parity coding below is classical in the stopping-time literature; see, for example, Terras [6]. Its short proof is included because no external result is needed by the argument.

Proposition 2.2 (parity-vector bijection)

For every $M \ge 0$,

$$n \bmod 2^M \longmapsto (p_0(n), \ldots, p_{M-1}(n)) \tag{2.5}$$

is a bijection from $\mathbb{Z}/2^M\mathbb{Z}$ to $\{0,1\}^M$.

Proof

The relation

$$n \equiv n' \pmod{2^{k+1}} \quad \Longrightarrow \quad T(n) \equiv T(n') \pmod{2^k}$$

shows inductively that the first M parity bits depend only on $n \bmod 2^M$. For the converse, induct on the word length. Suppose the first bit is e, and use the induction hypothesis to recover the residue $m = T(n) \bmod 2^k$ from the remaining k bits. If $e = 0$, the unique compatible residue is

$$n \equiv 2m \pmod{2^{k+1}}.$$

If $e = 1$, it is

$$n \equiv (2m-1)3^{-1} \pmod{2^{k+1}},$$

where 3 is a unit modulo 2^{k+1}. The first residue is even, the second is odd, and in either case applying T recovers $m \bmod 2^k$. Thus every word has exactly one preimage residue. □

Proposition 2.3 (exact affine iterate)

For every $n \geq 1$ and $k \geq 0$,

$$T^k(n) = \frac{3^{s_k(n)}}{2^k} n + \sum_{i=0}^{k-1} \frac{p_i(n) 3^{s_k(n) - s_{i+1}(n)}}{2^{k-i}}. \tag{2.6}$$

Proof

The assertion is trivial at $k = 0$. Passing from k to $k+1$, an even letter divides every existing term by two. An odd letter multiplies every existing term by $3/2$ and adds $1/2$. This is exactly the displayed update. □

3. A dense maximal-barrier set

The parity bijection identifies a complete dyadic shell with the Boolean cube. We now discard the sparse words whose prefixes deviate too far from their mean and obtain the pathwise orbit envelope needed to guarantee the first passages used later.

For a parity word of length M, put

$$Y_k = 2s_k - k, \qquad H_M = \max_{0 \leq k \leq M} \left| s_k - \frac{k}{2} \right| = \frac{1}{2} \max_{0 \leq k \leq M} |Y_k|. \tag{3.1}$$

For $0 \leq t < 1/2$, put

$$I(t) = D\left(\frac{1}{2} + t \,\middle\|\, \frac{1}{2}\right) = \left(\frac{1}{2} + t\right) \log(1+2t) + \left(\frac{1}{2} - t\right) \log(1-2t). \tag{3.2}$$

Lemma 3.1 (entropy-sharp maximal Boolean-walk bound)

If $M \geq 1$ and $0 \leq h < M/2$, then

$$2^{-M} \#\{w \in \{0,1\}^M : H_M(w) > h\} \leq 2\exp(-M I(h/M)). \tag{3.3}$$

In particular, the right side is at most $2\exp(-2h^2/M)$.

Moreover, if $0 < t_0 \le t \le t_1 < 1/2$, then there is a constant C_{t_0,t_1} such that, for every $M \ge 1$,

$$2^{-M}\#\{w : H_M(w) > tM\} \le C_{t_0,t_1} M^{-1/2} e^{-MI(t)}. \tag{3.3a}$$

Proof

Fix $\theta > 0$ and stop the Boolean tree at the first prefix u with $|Y(u)| > 2h$. For a frontier node of depth k, set

$$\Phi_\theta(u) = 2^{-k}\cosh(\theta Y(u))(\cosh\theta)^{M-k}.$$

The identity

$$\cosh(\theta(y-1)) + \cosh(\theta(y+1)) = 2\cosh(\theta y)\cosh\theta$$

shows that replacing an unexpanded node by its two children preserves the total potential. The frontier formed by first-crossing prefixes and noncrossing depth-M words therefore has total $(\cosh\theta)^M$. Each crossing prefix contributes at least $2^{-|u|}\cosh(2\theta h)$, whence

$$\Pr(H_M > h) \le 2\exp(M\log\cosh\theta - 2\theta h). \tag{3.4}$$

Take $u = 2h/M$ and $\theta = \operatorname{arctanh} u$. The elementary Legendre identity

$$u\theta - \log\cosh\theta = D\Big(\frac{1+u}{2} \,\Big\|\, \frac{1}{2}\Big) = I(h/M)$$

proves (3.3). Pinsker's inequality in this binary case, or the direct convexity estimate $I(t) \ge 2t^2$, gives the quadratic assertion.

It remains to prove (3.3a). Put $a = \lfloor 2tM \rfloor + 1$. Reflection, followed by symmetry between the two boundary signs, gives

$$\Pr(H_M > tM) \le 4\Pr(Y_M \ge a-1). \tag{3.3b}$$

This form is insensitive to the parity of a: replacing the terminal threshold by the next reachable lattice point only decreases the event. If

$$k_0 = \Big\lceil \frac{M+a-1}{2} \Big\rceil, \qquad p_M = \frac{k_0}{M},$$

then the probability on the right of (3.3b) is $2^{-M}\sum_{k=k_0}^{M}\binom{M}{k}$. For $k \ge k_0$, consecutive terms have ratio

$$\frac{\binom{M}{k+1}}{\binom{M}{k}} = \frac{M-k}{k+1}.$$

Uniformly for $t_0 \le t \le t_1$, this ratio is at most a fixed number strictly below one once M exceeds a fixed startup. The tail is therefore at most a fixed multiple of its first term. Uniform Stirling inequalities, with p_M confined to a compact subset of $(0,1)$, give

$$2^{-M}\binom{M}{k_0} \le CM^{-1/2}\exp\{-MD(p_M \,\|\, 1/2)\}.$$

Here $p_M = 1/2 + t + O(M^{-1})$, uniformly in the displayed compact interval. The derivative of $D(p\,\|\,1/2)$ is bounded on a slightly larger compact interval, so

$$MD(p_M \,\|\, 1/2) = MI(t) + O(1).$$

Absorbing this bounded error and the finite startup into C_{t_0,t_1} proves (3.3a). □

Set

$$\varrho := \frac{\sqrt{3}}{2} = 2^{a_0 - 1}.$$

Define

$$r_k(n) = \sum_{i=0}^{k-1} \frac{p_i(n)}{3^{s_{i+1}(n)} 2^{k-i}}, \qquad d_k = s_k - \frac{k}{2}. \tag{3.5}$$

By Proposition 2.3,

$$T^k(n) = \varrho^k n 3^{d_k} + (r_k(n) 3^{k/2}) 3^{d_k}. \tag{3.6}$$

Lemma 3.2 (uniform affine correction)

If $H_M \le h$, then for every $1 \le k \le M$,

$$r_k(n) 3^{k/2} \le (2 + \sqrt{3})(1 - \varrho^k) 3^h < (2 + \sqrt{3}) 3^h. \tag{3.7}$$

Proof

For $0 \le i < k$, the barrier gives $s_{i+1} \ge (i+1)/2 - h$. Hence

$$r_k 3^{k/2} \le 3^h \sum_{i=0}^{k-1} \frac{3^{(k-i-1)/2}}{2^{k-i}} = \frac{3^h}{2} \sum_{j=0}^{k-1} \varrho^j,$$

which is (3.7). □

For $0 < \eta \le 1$, let W_η be the set of integers n such that

$$\varrho^k n^{1-\eta} \le T^k(n) \le \varrho^k n^{1+\eta} \qquad (0 \le k \le \lfloor \log_2 n \rfloor). \tag{3.8}$$

Define

$$b_{\mathrm{ent}}(\eta) = I\left(\frac{\eta}{\log_2 3}\right). \tag{3.9}$$

Proposition 3.3 (entropy-sharp density of the maximal-barrier set)

Let $0 < \eta < 1 - a_0$. For every fixed $0 < c < b_{\mathrm{ent}}(\eta)$, there is $K_{\eta,c} > 0$ such that

$$\frac{\#(W_\eta^c \cap I_M)}{2^M} \le K_{\eta,c} e^{-cM} \qquad (M \ge 0). \tag{3.10}$$

Consequently W_η is $(K'_{\eta,c}, c/\log 2)$-dense after increasing the fixed constant.

Proof

Put $L_3 = \log_2 3$. Since $c < b_{\mathrm{ent}}(\eta)$, choose a fixed $0 < \lambda < 1$ with

$$c < I\left(\frac{\lambda\eta}{L_3}\right). \tag{3.11}$$

On I_M, take

$$h = \frac{\lambda\eta M}{L_3}. \tag{3.12}$$

Assume $H_M \le h$. Since $3^h = 2^{\lambda\eta M} \le n^\eta$, the multiplicative term in (3.6) is at least $\varrho^k n^{1-\eta}$ and at most

$$2^{-(1-\lambda)\eta M}\varrho^k n^{1+\eta}. \tag{3.13}$$

Lemma 3.2 bounds the additive term by

$$(2+\sqrt{3})3^{2h} = (2+\sqrt{3})2^{2\lambda\eta M}. \tag{3.14}$$

The unused part of the upper envelope is at least

$$\left(1-2^{-(1-\lambda)\eta M}\right)\varrho^M 2^{(1+\eta)M}. \tag{3.15}$$

Indeed, before this uniform replacement the slack is $\left(1-2^{-(1-\lambda)\eta M}\right)\varrho^k n^{1+\eta}$. Over all $0 \le k \le M$ and $n \in I_M$, its minimum occurs at $k = M$ and at the lower shell endpoint $n = 2^M$, which gives (3.15). Its binary exponent exceeds that of (3.14), because

$$a_0 + \eta - 2\lambda\eta \ge a_0 - \eta > 0. \tag{3.16}$$

Thus (3.8) holds for every $k \le M$ once M is sufficiently large.

Proposition 2.2 and Lemma 3.1 now give

$$\frac{\#(W_\eta^c \cap I_M)}{2^M} \le 2\exp\left[-MI\left(\frac{\lambda\eta}{L_3}\right)\right] \le 2e^{-cM}$$

on all sufficiently large shells. Enlarge $K_{\eta,c}$ to absorb the finite startup. Summing the resulting geometric shell series proves the global density assertion. □

4. First-passage linear transport

The barrier controls the retained sources, but re-certification also requires transporting an arbitrary bad landing set back to the original shell. First-passage reversal and loss-filtered fibers provide that bridge.

For real $Y > 1$, define

$$\tau_Y(n) = \min\{h \ge 0 : T^h(n) \le Y\} \tag{4.1}$$

when the set is nonempty.

Lemma 4.1 (first-passage band and reverse product)

Let $1 < Y < 2^M$, let $n \in I_M$, and suppose $\tau_Y(n) = h \ge 1$. Write $x_j = T^j(n)$, $y = x_h$, and let s be the number of odd values among $x_0, \dots, x_{h-1}$. Then

$$\frac{Y}{2} < y \le Y \tag{4.2}$$

and

$$n = \frac{2^h y}{3^s} \prod_{\substack{0 \le j < h \\ x_j \text{ odd}}} \left(1 - \frac{1}{2x_{j+1}}\right). \tag{4.3}$$

Consequently, whenever $h < 2Y$,

$$\left(1 - \frac{h}{2Y}\right)\frac{2^h y}{3^s} \le n \le \frac{2^h y}{3^s}. \tag{4.4}$$

Proof

The final crossing cannot be odd: otherwise $x_h = (3x_{h-1} + 1)/2 > x_{h-1} > Y$. Hence $x_{h-1} = 2y$, proving (4.2). Reversing an even step gives $x_j = 2x_{j+1}$, while reversing an odd step gives

$$x_j = \frac{2x_{j+1} - 1}{3} = \frac{2}{3}x_{j+1}\left(1 - \frac{1}{2x_{j+1}}\right).$$

Multiplication proves (4.3). Every odd factor occurs before the final crossing, so its following state is greater than Y. Therefore $\prod_i(1 - u_i) \ge 1 - \sum_i u_i$ gives a lower product bound $1 - s/(2Y) \ge 1 - h/(2Y)$; the upper bound is one. □

Write the reverse product in (4.3) as $\prod_{j<h}(1 - u_j)$, where

$$u_j(n) = \begin{cases} \dfrac{1}{2T^{j+1}(n)}, & T^j(n) \text{ odd}, \\ 0, & T^j(n) \text{ even}, \end{cases} \qquad E_Y(n) = Y\sum_{j=0}^{h-1} u_j(n). \tag{4.5}$$

The factor Y records the additive reverse loss at the target scale: E_Y/Y is an additive upper bound for the defect of the reverse product. It also makes concatenation compatible with changing thresholds. If a segment is measured locally at $Y' \ge Y$, then its contribution after rescaling to the final threshold Y is exactly Y/Y' times its local contribution to $E_{Y'}$. Thus segment losses add after rescaling, which is the coordinate used in Lemma 5.2. The elementary product inequality gives

$$1 - \frac{E_Y(n)}{Y} \le \prod_{j=0}^{h-1}(1 - u_j(n)) \le 1. \tag{4.6}$$

Lemma 4.2 (loss-filtered tagged fibers)

Let $D \ge 0$, and suppose

$$\frac{D}{Y} \le \frac{1}{3}. \tag{4.7}$$

For a fixed tag (h, y), the number of sources $n \in I_M$ satisfying

$$\tau_Y(n) = h, \qquad T^h(n) = y, \qquad E_Y(n) \le D$$

is at most

$$1 + 3D\frac{2^M}{Y}. \tag{4.8}$$

Proof

Put $P(n) = \prod_{j<h}(1 - u_j(n))$. By Lemma 4.1, if two such sources had odd counts $s_1 < s_2$, put $A_i = 2^h y/3^{s_i}$. Then $A_1 \ge 3A_2$, while (4.3) and (4.6) give

$$n_1 = A_1 P(n_1) \ge 3A_2\left(1 - \frac{D}{Y}\right) \ge 2A_2 \ge 2n_2.$$

This is impossible in the half-open shell I_M. The argument uses only positive products and introduces no division by a reverse product. Fix the common odd count and put $A = 2^h y/3^s$. All sources lie in $[(1 - D/Y)A, A]$. Nonemptiness, (4.7), and $n < 2^{M+1}$ imply $A < (3/2)2^{M+1}$, so this interval has length less than

$$3D\frac{2^M}{Y}.$$

This proves (4.8). □

Proposition 4.3 (loss-filtered target transport)

Put $J_Y = (Y/2, Y] \cap \mathbb{N}$. Let $1 < Y < 2^M$. Under (4.7), every $B \subseteq J_Y$ satisfies the exact bound

$$\#\left\{n \in I_M : \begin{matrix} \tau_Y(n) \le H, & T^{\tau_Y(n)}(n) \in B, \\ E_Y(n) \le D \end{matrix}\right\} \le H\left(1 + 3D\frac{2^M}{Y}\right)|B|. \tag{4.9}$$

Consequently,

$$\#\left\{n \in I_M : \begin{matrix} \tau_Y(n) \le H, & T^{\tau_Y(n)}(n) \in B, \\ E_Y(n) \le D \end{matrix}\right\} \le H(1 + 3D)\frac{2^M}{Y}|B|. \tag{4.10}$$

The estimate remains valid after arbitrary source restriction.

Proof

Since $Y < 2^M$, every passage time is positive. Sum Lemma 4.2 over the at most $H|B|$ tags to obtain (4.9). The contribution $H|B|$ is at most $H(2^M/Y)|B|$, which gives (4.10). □

5. Nested first-passage re-certification

The transport theorem applies directly to a first passage from the original shell. The nested construction below shows that every first certification failure has exactly this form and controls the reverse loss accumulated before it.

Fix

$$a_0 < r < 1, \qquad 0 < \eta < r - a_0, \qquad 0 < c_\eta < b_{\mathrm{ent}}(\eta). \tag{5.1}$$

Increase a fixed startup rank $M_0 = M_0(r, \eta, c_\eta)$ whenever needed below. If $x \in W_\eta \cap I_m$, put $q = \lfloor rm \rfloor$. For all $m \ge M_0$,

$$T^m(x) \le \varrho^m x^{1+\eta} < 2^{(a_0+\eta)m+1+\eta} \le 2^q. \tag{5.2}$$

Thus $\tau_{2^q}(x) \le m$, and (3.8) bounds the whole block by $x^{1+\eta}$.

Fix integers $M \ge L \ge M_0$, and start with $n_0 = n \in I_M$. Put

$$m_i = \lfloor \log_2 n_i \rfloor, \qquad t_0 = 0.$$

If $m_i < L$, stop successfully. If $m_i \ge L$ but $n_i \notin W_\eta$, stop with a certification failure. Otherwise define

$$q_i = \lfloor r m_i \rfloor, \quad h_i = \tau_{2^{q_i}}(n_i), \quad n_{i+1} = T^{h_i}(n_i), \quad t_{i+1} = t_i + h_i. \tag{5.3}$$

The first-passage band gives

$$2^{q_i - 1} < n_{i+1} \le 2^{q_i}, \qquad m_{i+1} \in \{q_i - 1, q_i\}. \tag{5.4}$$

Since $q_i \le m_i - 1$,

$$m_{i+1} \le q_i \le r m_i, \qquad q_{i+1} \le q_i - 1 \tag{5.5}$$

whenever the next block is executed. Consequently

$$t_i \le \sum_{j<i} m_j \le M \sum_{j<i} r^j < \frac{M}{1 - r}. \tag{5.6}$$

Lemma 5.1 (nested direct first passage)

For every executed block i,

$$t_{i+1} = \tau_{2^{q_i}}(n_0), \qquad n_{i+1} = T^{\tau_{2^{q_i}}(n_0)}(n_0). \tag{5.7}$$

Proof

The assertion is the definition when $i = 0$. Inductively, every state before t_i exceeds the preceding threshold and hence the strictly smaller 2^{q_i}. At time t_i, one has $n_i \geq 2^{m_i} > 2^{q_i}$. The local block stays above 2^{q_i} until its endpoint. Thus $t_i + h_i$ is exactly the first crossing of the new threshold by the original orbit. □

For $q \geq 1$, put

$$J_q = (2^{q-1}, 2^q] \cap \mathbb{N}, \qquad B_q = W_\eta^c \cap J_q. \tag{5.8}$$

The interval J_q lies in I_{q-1} apart from its upper endpoint. Proposition 3.3 therefore gives

$$\frac{|B_q|}{2^q} \leq C_{\eta,c_\eta}(e^{-c_\eta q} + 2^{-q}). \tag{5.9}$$

Lemma 5.2 (rank-scaled reverse loss)

If blocks $0, \ldots, i$ are executed before the first certification failure, then the direct first passage in (5.7) satisfies

$$E_{2^{q_i}}(n) < \frac{q_i + 2}{r}. \tag{5.10}$$

Proof

In block j, every state following an odd step is strictly larger than 2^{q_j}; the final crossing is even. Therefore each reverse-loss increment is less than 2^{-q_j-1}. Since the block has at most m_j steps, rescaling all blocks to the final threshold gives

$$E_{2^{q_i}}(n) < \frac{1}{2} \sum_{j \leq i} m_j 2^{q_i - q_j}.$$

Now $m_j < (q_j + 1)/r$, and the distinct integer thresholds q_j are all at least q_i. Hence

$$E_{2^{q_i}}(n) < \frac{1}{2r} \sum_{d \geq 0} (q_i + d + 1) 2^{-d} = \frac{q_i + 2}{r}.$$

□

Let $\mathrm{Fail}_{M,L}$ denote the sources whose chain reaches a certification failure before it reaches rank below L.

Theorem 5.3 (optimized terminal profile)

Under (5.1), for the fixed startup rank M_0 chosen above there is a constant $C > 0$ such that, for all integers $M \geq L \geq M_0$,

$$\boxed{\frac{\#\,\mathrm{Fail}_{M,L}}{2^M} \leq C\left[e^{-c_\eta M} + M(L+1)(e^{-c_\eta L} + 2^{-L})\right].} \tag{5.11}$$

Every source outside this set has an integer $k < M/(1-r)$ such that

$$T^k(n) < 2^L, \qquad \max_{0 \leq j \leq k} T^j(n) \leq n^{1+\eta}. \tag{5.12}$$

Proof

An initial failure has proportion $O(e^{-c_\eta M})$. Otherwise let the first failed certification be a generated landing in B_q. Every earlier block and the block producing that landing are certified, so Lemma 5.2 applies; the bad landing itself is not assumed certified. Lemma 5.1 makes it a direct first passage from I_M, at time at most $M/(1-r)+1$.

Increase M_0, if necessary, so that

$$\frac{q+2}{r2^q} \le \frac{1}{3} \tag{5.13}$$

for every $q \ge M_0$. In Proposition 4.3, the available horizon satisfies $H \le M/(1-r)+1 \ll_r M$, while with $D=(q+2)/r$ one has $1+3D \ll_r q+1$. Combining (4.10) with (5.9) therefore bounds the proportion of sources whose first failed landing has rank q by

$$CM(q+1)(e^{-c_\eta q}+2^{-q}). \tag{5.14}$$

The two elementary tail estimates

$$\sum_{q\ge L}(q+1)e^{-c_\eta q} \ll (L+1)e^{-c_\eta L}, \qquad \sum_{q\ge L}(q+1)2^{-q} \ll (L+1)2^{-L}$$

prove (5.11).

Outside the failure set, integer ranks strictly decrease until a state of rank below L is reached. Equations (5.2), (5.6), and the monotonic decrease of block endpoints prove (5.12). □

6. Shrinking barriers and compressed time support

The loss in Theorem 5.3 comes from summing over all times up to a linear horizon. The actual certified blocks occupy a much smaller time support once the high-rank barrier width is allowed to shrink with the current rank.

Choose fixed high-rank data

$$a_0 < r_{hi} < 1, \qquad 0 < \tau < r_{hi} - a_0, \tag{6.1}$$

and fixed low-rank data

$$a_0 < r_{lo} < 1, \qquad 0 < \eta_{lo} < r_{lo} - a_0. \tag{6.2}$$

Put $r_* = \min\{r_{hi}, r_{lo}\}$. Fix L_{loss} so large that

$$\frac{q+2}{r_*2^q} \le \frac{1}{3} \qquad (q \ge L_{loss}).$$

For constants $D_{hi}, C_{sw} > 0$, put

$$S_M = \lceil C_{sw}\log(M+2)\rceil, \tag{6.3}$$

and, at a high parent rank $m \ge S_M$, certify with

$$\eta_{M,m} = \min\left\{\tau, D_{hi}\sqrt{\frac{\log(M+2)}{m}}\right\}. \tag{6.4}$$

Below the switch use the fixed tolerance η_{lo}. Let M_{hi} and M_{lo} be fixed startup ranks for the parameter pairs (r_{hi}, τ) and (r_{lo}, η_{lo}), respectively, as in Section 5, and put

$$L_0 = \max\{2, M_{hi}+1, M_{lo}+1, L_{loss}\}. \tag{6.4a}$$

The high-rank stage construction at tolerance τ remains valid after lowering its tolerance to $\eta_{M,m}$; the density of the resulting rank-dependent certification sets will be estimated separately below.

The chain below has the same stopping and first-failure logic as the chain in Section 5. Only the tolerance changes: it shrinks above the switch rank and is fixed below it.

Here is the literal stopped chain used in this section. Fix M, write $S = S_M$, and set

$$(r(m), t_M(m)) = \begin{cases} (r_{hi}, \eta_{M,m}), & m \geq S, \\ (r_{lo}, \eta_{lo}), & m < S. \end{cases} \tag{6.4b}$$

For a landing at threshold rank q, define the certification that would be used by its lower-shell continuation by

$$\vartheta_{M,S}(q) = \begin{cases} \eta_{M,q-1}, & S \leq q-1, \\ \eta_{lo}, & q-1 < S. \end{cases} \tag{6.4c}$$

Start with $n_0 = n \in I_M$, parent rank $m_0 = M$, and elapsed time $H_0 = 0$. If $n_0 \notin W_{\eta_{M,M}}$, stop with an initial certification failure. Otherwise, from a certified checkpoint $n_i \in I_{m_i}$, set

$$q_i = \lfloor r(m_i) m_i \rfloor, \qquad h_i = \tau_{2^{q_i}}(n_i), \qquad n_{i+1} = T^{h_i}(n_i), \qquad H_{i+1} = H_i + h_i. \tag{6.4d}$$

If $q_i < L$, stop successfully. If $q_i \geq L$ and $n_{i+1} \notin W_{\vartheta_{M,S}(q_i)}$, stop at the unique first failed landing, of rank q_i. Otherwise set $m_{i+1} = \lfloor \log_2 n_{i+1} \rfloor$ and continue from n_{i+1}.

Put

$$B^{sh}_{M,S,q} = ((2^{q-1}, 2^q] \cap \mathbb{N}) \setminus W_{\vartheta_{M,S}(q)}. \tag{6.4e}$$

Then $\mathrm{Fail}^{sh}_{M,L}$ is, by definition, the union of the initial failures in $I_M \setminus W_{\eta_{M,M}}$ and the sources $n \in I_M$ for which some executed block i ends at cumulative time H_{i+1} and rank $L \leq q_i < M$ with $T^{H_{i+1}}(n) \in B^{sh}_{M,S,q_i}$. This is a first-bad union: the bad landing is never used as the source of another certified block.

The nested first-passage proof remains unchanged. Every first-bad landing at threshold rank q is a direct first passage from the original shell and satisfies

$$E_{2^q}(n) < \frac{q+2}{r_*}. \tag{6.5}$$

Lemma 6.1 (duration corridor and feasible-time support)

There is a constant $K > 0$, depending only on the fixed parameters, such that for every outer shell I_M and every possible first-bad rank q, all cumulative first-passage times belong to a finite set $H_{M,q}$ satisfying

$$\# H_{M,q} \leq K \sqrt{(M+2)\log(M+2)}. \tag{6.6}$$

Proof

Write $g = 1 - a_0$. Consider one certified block from a parent shell I_m to threshold 2^q, and let h be its duration and t its active tolerance. The two deterministic orbit envelopes and the first-passage band give

$$(1-t)m - q \leq gh < (1+t)(m+1) - q + 1. \tag{6.7}$$

Thus

$$|gh - (m-q)| \leq tm + t + 2. \tag{6.7a}$$

Now suppose blocks $0, \ldots, j$ have been executed. Every landing before the last one is certified. Such a landing lies in $(2^{q_i - 1}, 2^{q_i}]$. It cannot equal 2^{q_i}: if it did, then at time q_i its certification lower bound would be

$$\varrho^{q_i}(2^{q_i})^{1-t} = 2^{(a_0-t)q_i} > 1 = T^{q_i}(2^{q_i}),$$

because every active tolerance satisfies $t < a_0$. Hence each certified landing lies in the unique shell I_{q_i-1}, and therefore

$$m_{i+1} = q_i - 1 \qquad (0 \le i < j). \tag{6.7b}$$

This removes the apparent rank branching. In particular,

$$\sum_{i=0}^{j} (m_i - q_i) = M - q_j - j. \tag{6.7c}$$

Summing (6.7a), using (6.7c), and paying one additional unit per block to change the center from $M - q_j - j$ to $M + 1 - q_j$, gives the exact cumulative corridor

$$|gH_{j+1} - ((M+1) - q_j)| \le \sum_{i=0}^{j} (t_i m_i + t_i + 3). \tag{6.7d}$$

For high blocks,

$$t_i m_i \le D_{hi}\sqrt{m_i \log(M+2)}, \qquad t_i \le \tau,$$

and (6.7b) gives $m_{i+1} \le r_{hi} m_i$. The resulting geometric square-root sum is $O(\sqrt{M\log(M+2)})$. The low phase starts below S_M, and its parent ranks decrease with ratio at most r_{lo}; hence its fixed-tolerance contribution is $O(S_M)$. Since $S_M = O(\log(M+2))$, the right side of (6.7d) is at most $K_0\sqrt{(M+2)\log(M+2)}$.

For fixed M and $q_j = q$, the center in (6.7d) is fixed. Consequently every feasible integer H_{j+1} lies in one interval of length $O(\sqrt{M\log(M+2)})$; counting its integer points proves (6.6). No union over intermediate rank histories remains. □

The tagged-fiber proof also has the following support-sensitive form. If H is any finite set of positive times, $B \subseteq [1, Y]$, and the scaled reverse loss is at most D_q, then

$$\#\left\{n \in I_M : \begin{array}{l}\text{the first passage below } Y \text{ occurs at some } h \in H,\\ T^h(n) \in B,\ E_Y(n) \le D_q\end{array}\right\} \le \#H\left(1 + 3D_q\frac{2^M}{Y}\right)\#B. \tag{6.8}$$

This is Proposition 4.3 summed only over the declared time tags. No interval structure or density hypothesis on H is used.

Proposition 6.2 (shrinking high-rank density)

Assume

$$\frac{D_{hi}}{\sqrt{C_{sw}}} \le \tau \qquad \text{and} \qquad 1 \le S_M \le M.$$

Put

$$c_0 = \frac{1}{2(\log_2 3)^2}, \qquad C_0 = 2e^{4c_0}.$$

Then the cap in (6.4) is inactive at every $m \ge S_M$, and

$$\eta_{M,m}^2 m = D_{hi}^2 \log(M+2). \tag{6.8a}$$

Moreover,

$$\frac{\#\,(I_M \setminus W_{\eta_{M,M}})}{2^M} \le C_0(M+2)^{-c_0 D_{hi}^2}, \tag{6.8b}$$

and, for every q with $S_M \le q-1$,

$$\frac{|B^{sh}_{M,S_M,q}|}{2^q} \le 2^{-q} + \frac{C_0}{2}(M+2)^{-c_0 D_{hi}^2}. \tag{6.8c}$$

Proof

From $m \ge S_M \ge C_{sw}\log(M+2)$,

$$D_{hi}\sqrt{\frac{\log(M+2)}{m}} \le \frac{D_{hi}}{\sqrt{C_{sw}}} \le \tau.$$

This proves that the cap is inactive and gives (6.8a).

For completeness, the uniform quadratic estimate used here is

$$\frac{\#\,(I_m \setminus W_t)}{2^m} \le C_0 e^{-c_0 t^2 m} \qquad (0 < t \le 1). \tag{6.8d}$$

When $m \ge 4$ and $tm \ge 2$, apply Lemma 3.1 at height $tm/(2\log_2 3)$ and use $I(u) \ge 2u^2$; the correction argument of Lemma 3.2 then gives the required orbit envelope. Outside this startup regime, $t^2 m \le 4$, so the trivial shell bound is absorbed by $C_0 = 2e^{4c_0}$. Thus (6.8d) is uniform in t. Applying it with $t = \eta_{M,M}$ and using (6.8a) proves (6.8b).

For $q \ge 1$, the landing band lies in I_{q-1} apart from its single upper endpoint 2^q. Apply (6.8d) in I_{q-1}, divide by 2^q, and use (6.8a) at $m = q-1$. This gives (6.8c). □

Consequently the initial high failure and every high-rank landing target have a common density bound

$$d_{hi}(M) \ll (M+2)^{-p_{hi}}, \qquad p_{hi} = \min\{C_{sw}\log 2, c_0 D_{hi}^2\}, \tag{6.9}$$

where $c_0 > 0$ is the fixed quadratic maximal-barrier constant. The first term in the minimum pays for the dyadic endpoint boundary at the switch.

Theorem 6.3 (support-sensitive terminal profile)

Let $0 < b_{lo} < b_{ent}(\eta_{lo})$ and let $0 < c_2 < \log 2$. Assume $D_{hi}/\sqrt{C_{sw}} \le \tau$. There is a fixed startup rank $M_1 \ge L_0$, depending only on the displayed parameters, such that, for all integers

$$M \ge M_1, \qquad L_0 \le L < S_M < M,$$

$$\boxed{\frac{\#\,\mathrm{Fail}^{sh}_{M,L}}{2^M} \ll d_{hi}(M) + \sqrt{M\log(M+2)}\,(M^2 d_{hi}(M) + e^{-b_{lo}L} + e^{-c_2 L}).} \tag{6.10}$$

Outside this failure set, some iterate before

$$\frac{M}{1-r_{hi}} + \frac{S_M+1}{1-r_{lo}}$$

lies below 2^L. If $\tau < \beta$, every iterate through the same witness is at most $n^{1+\beta}$, once M is sufficiently large.

Proof

Choose an intermediate rate $b_{lo} < \tilde{b}_{lo} < b_{ent}(\eta_{lo})$, and increase M_1 until all fixed barrier startups, the switch inequalities, and the loss-filter condition above (6.3) hold.

Partition by the unique first failed certification rank. The direct-passage identity (6.5), Lemma 6.1, and (6.8) give, after division by 2^M,

$$\frac{\#\{\text{first failure at rank } q\}}{2^M} \ll \sqrt{M\log(M+2)}\,(q+1)\frac{|B^{\mathrm{sh}}_{M,S_M,q}|}{2^q}. \tag{6.10a}$$

If $q \geq S_M + 1$, then (6.4c) selects $\eta_{M,q-1}$, including the possible upper endpoint 2^q, and Proposition 6.2 bounds the last factor by $d_{\mathrm{hi}}(M)$. Hence

$$\sum_{q=S_M+1}^{M-1} (q+1)d_{\mathrm{hi}}(M) \ll M^2 d_{\mathrm{hi}}(M). \tag{6.10b}$$

If $L \leq q \leq S_M$, then (6.4c) selects the fixed tolerance η_{lo}. Proposition 3.3, applied in I_{q-1} and with the single upper endpoint retained, gives

$$\frac{|B^{\mathrm{sh}}_{M,S_M,q}|}{2^q} \ll e^{-\tilde{b}_{\mathrm{lo}}q} + 2^{-q}.$$

The two strict rate gaps therefore give

$$\sum_{q=L}^{S_M} (q+1)\frac{|B^{\mathrm{sh}}_{M,S_M,q}|}{2^q} \ll e^{-b_{\mathrm{lo}}L} + e^{-c_2 L}. \tag{6.10c}$$

Combining (6.10a)–(6.10c) yields the transported terms in (6.10). The initial failure contributes $d_{\mathrm{hi}}(M)$ by (6.8b).

The block ranks decrease geometrically, which gives the displayed clock. High blocks use tolerance at most τ. Every low block begins below $2^{S_M+1} = (M+2)^{O(1)} = n^{o(1)}$, so its fixed-tolerance envelope is also below $n^{1+\beta}$ eventually. □

Critical moving low-rank profile

The fixed low tolerance in Theorem 6.3 proves every strict exponent above A_{FP}. To reach its principal endpoint, the low parameters must approach their entropy boundary at the same rate as the terminal rank. Put

$$\eta_* = 1 - a_0.$$

Let $L = L_M \asymp \log M$, choose fixed sufficiently large constants K_0, K_1, and use throughout the low part of the M-th chain

$$\eta_M = \eta_* - \frac{K_0}{L}, \qquad r_M = 1 - \frac{K_0}{2L}, \qquad \lambda_M = 1 - \frac{K_1}{L}. \tag{6.11}$$

For every low parent rank $m \geq L$,

$$(r_M - a_0 - \eta_M)m = \frac{K_0 m}{2L} \geq \frac{K_0}{2}, \qquad (1-\lambda_M)\eta_M m \geq K_1\eta_M. \tag{6.12}$$

Choose K_0 large enough to absorb the floor in $q = \lfloor r_M m \rfloor$, the upper endpoint of the parent shell, and the fixed affine startup constants. The first inequality then guarantees passage below 2^q by time m. The second leaves a uniform positive fraction of the multiplicative upper envelope. Finally,

$$a_0 + \eta_M - 2\lambda_M\eta_M \longrightarrow a_0 - \eta_* = 2a_0 - 1 > 0. \tag{6.13}$$

Thus, after one fixed startup, the additive correction is absorbed uniformly for every $m \geq L$. This proves the same all-prefix certification and first-passage conclusions as the fixed low pair, now with (r_M, η_M).

For precision, the moving chain is the stopped chain (6.4b)–(6.4e) with the high branch unchanged and with (r_{lo}, η_{lo}) replaced throughout the low branch by (r_M, η_M). Thus a low landing of threshold rank q is tested against W_{η_M}, and we write

$$B^{crit}_{M,q} = ((2^{q-1}, 2^q] \cap N) \setminus W_{\eta_M}.$$

The failure set $\mathrm{Fail}^{crit}_{M,L}$ is the corresponding initial failure and first-bad union. As before, a bad landing is never used as the source of a later block. These definitions make the direct-passage identity and the loss estimate (6.5) applicable without changing their proofs.

Set

$$b_M = I\left(\frac{\lambda_M \eta_M}{\log_2 3}\right), \qquad b_* = I\left(\frac{\eta_*}{\log_2 3}\right) = \varkappa_* \log 2. \tag{6.14}$$

The arguments of I remain in a fixed compact subset of $(0, 1/2)$. Its derivative is bounded there, so

$$b_M \geq b_* - \frac{C}{L}. \tag{6.15}$$

Apply the sharp-prefactor clause (3.3a) on the landing shell I_{q-1}, retaining its single upper endpoint. Since $L \leq q \leq S_M = O(\log M)$ and $L \asymp \log M$, the factor $\exp(Cq/L)$ is bounded. Therefore the low first-bad target satisfies

$$\boxed{\frac{|B^{crit}_{M,q}|}{2^q} \ll q^{-1/2} 2^{-\varkappa_* q} + 2^{-q}} \qquad (L \leq q \leq S_M). \tag{6.16}$$

The change $r_M \to 1$ requires one modification to the time-support proof. The high phase is unchanged. In the low phase, strict integer threshold descent gives at most S_M blocks; every low parent rank and every low block duration is at most S_M. Each term in the right side of the cumulative corridor (6.7d) is also $O(S_M)$. Hence both the low elapsed time and its contribution to the corridor width are

$$O(S_M^2) = O((\log M)^2) = o(\sqrt{M \log M}). \tag{6.17}$$

Consequently Lemma 6.1 retains its $O(\sqrt{M \log M})$ conclusion for the moving low schedule. Eventually $r_M > r_{hi}$, so the fixed rank-scaled loss denominator remains r_{hi}.

Proposition 6.4 (critical rank-buffer profile)

Let $L_M \asymp \log M$ be an integer terminal-rank sequence. For every fixed $c > c_*$ and $\beta > 0$, the high parameters and switch constant may be chosen so that the moving low chain above has, for some fixed $\varepsilon > 0$,

$$\boxed{\frac{\#\,\mathrm{Fail}^{crit}_{M,L_M}}{2^M} \ll \sqrt{M \log M}\,(L_M^{1/2} 2^{-\varkappa_* L_M} + L_M 2^{-L_M}) + M^{-\varepsilon}.} \tag{6.18}$$

Outside this failure set, some iterate before $c \log n$ lies below 2^{L_M}, and every iterate through the same witness is at most $n^{1+\beta}$.

Proof

Choose

$$a_0 < r_{hi} < 1 - \frac{1}{c \log 2}, \qquad 0 < \tau < \min\{r_{hi} - a_0, \beta, a_0\}. \tag{6.19}$$

Because $L_M = O(\log M)$, choose C_{sw} so that $L_M < S_M$ eventually. Increase D_{hi} and then C_{sw}, preserving $D_{hi}/\sqrt{C_{sw}} \le \tau$, until the high contribution in (6.10) is $O(M^{-\varepsilon})$ with a fixed positive margin.

For a low first-bad rank, the direct-passage identity, the retained feasible time support, and loss-filtered transport give the rankwise estimate (6.10a), with $B^{crit}_{M,q}$ in place of the fixed low target. Equation (6.16) and elementary geometric-tail summation give

$$\sum_{q=L_M}^{S_M} (q+1)\frac{|B^{crit}_{M,q}|}{2^q} \ll L_M^{1/2} 2^{-\varkappa_* L_M} + L_M 2^{-L_M}. \tag{6.20}$$

Combining the low sum with the high contribution proves (6.18).

The high-phase clock is at most $M/(1-r_{hi}) + o(M)$. The low phase costs only $O((\log M)^2) = o(M)$ by (6.17); hence the total is less than $cM \log 2 \le c \log n$ eventually. The high blocks use tolerance at most $\tau < \beta$, while the low phase starts at polylogarithmic height. The same deterministic envelopes therefore retain the stated orbit ceiling. □

For later use, define the exact rank buffer

$$\Delta_M = \varkappa_* L_M - \frac{1}{2}\log_2(M+2) - \log_2 \log(M+3). \tag{6.21}$$

If $L_M \asymp \log M$, then the first term in (6.18) is $O(2^{-\Delta_M})$. If additionally $\Delta_M \to \infty$, then $L_M \ge (A_{FP} + o(1)) \log_2 M$; since $A_{FP} > 1/2$, the term containing 2^{-L_M} is absorbed into a fixed negative power of M. Thus

$$\boxed{\frac{\#\ \mathrm{Fail}^{crit}_{M,L_M}}{2^M} \ll 2^{-\Delta_M} + M^{-\varepsilon}.} \tag{6.22}$$

Proof of Theorem 1.1

For the bounded profile (A_M), use the terminal rank in (1.3). Condition (1.4) implies $L_M \asymp \log M$: the upper bound follows from boundedness, and the lower bound follows from the definition of Δ_M. Proposition 6.4 and (6.22) therefore give the shellwise estimate (1.5).

The same condition makes A_M eventually positive. Since the profile is bounded and $n \in I_M$,

$$2^{L_M} \le 2(M+2)^{A_M} \ll (\log n)^{A_M}, \tag{6.23}$$

with a constant uniform in M. This converts the terminal rank to the stated landing. Finally, the right side of (1.5) tends to zero, so the qualitative part of Lemma 2.1 assembles the shell good sets into one natural-density-one set. □

Proof of Corollary 1.2

For fixed $A > A_{FP}$, take $A_M = A$. Then

$$2^{-\Delta_M} \ll M^{-\varkappa_*(A-A_{FP})} \log M.$$

The high exponent in Proposition 6.4 can be chosen larger than any prescribed $\gamma < \varkappa_*(A - A_{FP})$. Splitting the dyadic shell sum at half the top rank gives the first assertion.

For the second assertion take

$$L_M = \lceil A_{FP} \log_2(M+2) + B \log_2 \log(M+3) \rceil.$$

Then $2^{-\Delta_M} \ll (\log M)^{1-B\varkappa_*}$. The same dyadic split gives every $\gamma < B\varkappa_* - 1$.

For the third assertion take

$$L_M = \lceil A_{FP} \log_2(M+2) + \frac{1}{\varkappa_*}\log_2 \log(M+3) + D \log_2 \log\log(M+4) \rceil. \tag{6.24}$$

Then $2^{-\Delta_M} \ll (\log\log M)^{-D\varkappa_*}$, and the shell split gives every $\gamma < D\varkappa_*$. The three displayed targets follow by exponentiating their terminal ranks and using $M \asymp \log n$.

For the final functional statement, let $\Omega(x) \to \infty$. Choose numbers $x_j \uparrow \infty$ so rapidly that $\Omega(x) \geq j$ for every $x \geq x_j$ and $\log j / \log x_j \to 0$. The step function equal to j on $[x_j, x_{j+1})$ is an eventually nondecreasing subpower minorant $\tilde{\Omega} \leq \Omega$, after a harmless fixed shift of its argument. Use (6.24) with the final term replaced by $\log_2 \tilde{\Omega}(\log(M+4))$. Its rank buffer tends to infinity, and the resulting smaller target implies the asserted Ω-target. □

7. Quantitative companions

The terminal profiles have different quantitative consumers. The support-sensitive profile gives every strict fixed polylogarithmic exponent, and its moving refinement Proposition 6.4 gives the endpoint profile in the headline. The one-regime profile gives the sharper stretched-logarithmic exceptional rate and the clock consequences recorded here.

Proof of Theorem 1.3

We prove the target in (1.8) by choosing the terminal rank so that $2^{L_M} \leq \exp((\log n)^{1-\delta})$ on the source shell.

Fix $0 < \delta < 1$, put $\alpha = 1 - \delta$, and choose a one-regime pair (r, η) satisfying (5.1), with

$$r < 1 - \frac{1}{c \log 2}, \qquad \eta < \beta. \tag{7.1}$$

Take

$$L_M = \lfloor \frac{(M \log 2)^\alpha}{\log 2} \rfloor. \tag{7.2}$$

The factor $M(L_M + 1)$ in (5.11) is absorbed by the exponential tail, so

$$\# \operatorname{Fail}_{M, L_M} \leq C 2^M e^{-c' M^\alpha}. \tag{7.3}$$

For $n \in I_M$,

$$2^{L_M} \leq \exp((M \log 2)^\alpha) \leq \exp((\log n)^\alpha).$$

The clock and path ceiling follow from (5.12) and (7.1). Lemma 2.1, with $\sigma = \alpha$, proves (1.8). □

Raw-clock conversion

Let $s_h(x)$ count the odd states among $x, T(x), \ldots, T^{h-1}(x)$. Direct induction gives

$$\operatorname{Col}^{h + s_h(x)}(x) = T^h(x). \tag{7.4}$$

If $x \in W_\eta$ and $h \leq \lfloor \log_2 x \rfloor$, the positive main term in the affine iterate and the upper envelope in (3.8) give

$$s_h(x) \leq \frac{h}{2} + \frac{\eta \log x}{\log 3}. \tag{7.5}$$

Thus a one-regime chain has raw time at most

$$\frac{1}{1-r}\left(\frac{3}{2\log 2} + \frac{\eta}{\log 3}\right) \log n + o(\log n). \tag{7.6}$$

For the fixed-low two-regime chain, (7.6) applies to the high phase and the low phase has $O(S_M)$ shortcut steps. For the moving endpoint, the low phase has $O(S_M^2)$ shortcut steps by (6.17). Both are $o(\log n)$ and incur at most twice as many raw steps. As $r \downarrow a_0$ and $\eta \downarrow 0$, the coefficient in (7.6) tends to

$$\frac{3}{\log(4/3)}.$$

The high parameters in (6.19) can therefore be chosen to satisfy any fixed shortcut and raw budgets strictly above their respective limits. To retain the literal raw-orbit ceiling in Corollary 1.4, run the argument with an internal exponent $0 < \beta' < \beta$. Every additional raw odd image is $3m + 1 = 2T(m)$, so $2n^{1+\beta'} \le n^{1+\beta}$ eventually. This proves Corollary 1.4. □

Fixed-power exceptional count

Fix $\alpha > 0$ and $0 < \sigma < 1$. Choose $0 < \delta < 1 - \sigma$. Then

$$\exp((\log n)^{1-\delta}) \le n^{\alpha}$$

eventually, while $1 - \delta > \sigma$. Theorem 1.3 therefore implies (1.10), after decreasing the positive exponential constant and absorbing the finite startup.

The independent fixed-depth argument for the graded clock (1.11) is given in Appendix A. It is not an input to either headline theorem.

8. Scope

The argument proves an orbit-minimum statement on a set of natural density one. It does not prove descent for every starting value, exclude nontrivial cycles, or control the orbit after the selected witness. The endpoint $\delta = 1$ is a strict parameter limit and is not claimed. No finite computation is used as an all-depth premise. The Collatz conjecture predicts $T_{\min}(n) = 1$ for every positive starting value; that universal conclusion is not proved here.

At the polylogarithmic endpoint, the proof requires $\Delta_M \to \infty$. It therefore does not prove the pure target $C(\log n)^{A_{FP}}$, nor the critical secondary target with a bounded tertiary multiplier. The moving theorem improves the terminal scale while weakening its displayed exceptional-set rate; these are separate comparison axes, as recorded in Corollary 1.2.

The proofs of Theorems 1.1 and 1.3 use only the optimized first-passage chain: the parity-vector bijection, maximal-barrier estimate, first-passage reversal, exact reverse-product loss, nested direct re-certification, and compressed feasible-time support. These proofs use no fixed-time endpoint-fiber moment, Fourier estimate, stochastic independence between orbit blocks, mixing statement, Diophantine reduction, or generated-target equidistribution hypothesis. The Boolean walk in Lemma 3.1 is exact counting over the parity words supplied by Proposition 2.2, not a stochastic model of an individual Collatz orbit. The independent graded-clock companion uses only a fixed finite dense-set pullback and is not on either headline dependency chain.

Research and software disclosure

Generative-AI systems were used in proof exploration, adversarial auditing, finite diagnostics, formalization, and exposition. The author selected the theorem and proof architecture, reviewed the resulting artifacts, and accepts responsibility for the manuscript. Finite diagnostics are supporting evidence only and are not premises of any theorem.

The separate Lean package formalizes the fixed-parameter optimized chain, from loss-filtered transport through the natural-density assembly, and separately formalizes the fixed-depth graded-clock companion. Its public `Main` theorem has the strict ranges $A > A_{FP}$, $c > c_*$, and $\beta > 0$, matching the fixed-A landing, clock, ceiling, and a positive logarithmic exceptional exponent in Corollary 1.2, part 1; its landing inequality is strict and hence slightly stronger than the manuscript's weak inequality. The exact upper range for that exceptional exponent is paper-level. For Theorem 1.3, Lean formalizes the literal landing and ceiling together with every strict exceptional power below $1 - \delta$; the endpoint power $1 - \delta$ in (1.8) is paper-level. The sharp binomial prefactor (3.3a), the moving low parameters (6.11), and the moving endpoint theorem are proved in the manuscript but are not yet represented by a public Lean theorem. Accordingly the formal artifact is evidence for the fixed-exponent specialization, not a machine-checked proof of the new moving endpoint refinement.

The frozen fixed-exponent formal snapshot is the access-controlled GitHub repository at commit `e382a241e73ac2f0a958b7411f7778584f3dc48d`; repository access can be provided to referees on request. That commit is the artifact whose narrower formal scope is described above; the V3.1 manuscript refinement will require a new immutable snapshot after its formalization boundary is accepted. The formal package uses Lean `4.15.0` (commit `11651562caae`) and Mathlib revision `9837ca9d65d9de6fad1ef4381750ca688774e608`. From the repository's `lean/` directory, the referee-facing build command is `lake build FirstPassageLinearTransport.PaperDependencyAudit FirstPassageLinearTransport.PaperAudit FirstPassageLinearTransport.Main`. The theorem dictionary is `lean/FORMALIZATION.md`; the dependency and axiom reports are respectively `lean/FirstPassageLinearTransport/PaperDependencyAudit.lean` and `lean/FirstPassageLinearTransport/PaperAudit.lean`. The build, placeholder scan, dependency report, and public-root axiom audit pass, with logical dependencies `propext`, `Classical.choice`, and `Quot.sound`. These checks supplement rather than replace the written proof.

Appendix A. Graded fixed-power clock

This appendix proves the second assertion of Corollary 1.5. The argument is a fixed-depth companion to the headline chain and records a time–descent tradeoff that the stronger polylogarithmic target alone does not express.

Put $g = 1 - a_0$. Fix $a_0 < r < 1$ and $0 < \eta < r - a_0$. For a certified source $x \in I_m$, set $Y_m = 2^{\lfloor rm \rfloor}$, let $G(x)$ be its first entrance into $(Y_m/2, Y_m]$, and put

$$H_m = \left\lceil \frac{(1+\eta-r)m + 2 + \eta}{g} \right\rceil. \tag{A.1}$$

The strict inequality $\eta < r - a_0$ gives $H_m \le m$ eventually. The upper half of the all-prefix envelope (3.8) then gives

$$T^{H_m}(x) < 2^{-gH_m + (1+\eta)(m+1)} \le 2^{rm-1} \le Y_m. \tag{A.2}$$

Hence the first passage defining G occurs by time H_m, and

$$H_m \le \frac{1+\eta-r}{g} m + O_{r,\eta}(1). \tag{A.3}$$

We record why a fixed number of these stages retains natural density one. After absorbing finitely many startup shells, let $\tilde{W}_\eta$ denote the certified set and put $P(S) = \tilde{W}_\eta \cap G^{-1}(S)$. If S is (C, D_{dens})-dense and $0 < \chi < r$, then $P(S)$ is $(C', \chi D_{\mathrm{dens}})$-dense for every sufficiently small fixed $D_{\mathrm{dens}} > 0$. Indeed, for every first passage of length at most H_m, the loss in (4.5) satisfies $E_{Y_m} \le D_{\mathrm{loss}} := H_m/2$, and $H_m/(2Y_m) \le 1/3$ eventually. Apply Proposition 4.3 with $D = D_{\mathrm{loss}}$ and with $B = (Y_m/2, Y_m] \setminus S$. Since $|B| \le C Y_m^{1-D_{\mathrm{dens}}}$, the transported part of the shell complement is

$$\ll C m^2 \frac{2^m}{Y_m} Y_m^{1-D_{\mathrm{dens}}} \ll C m^2 2^{(1-rD_{\mathrm{dens}})m}. \tag{A.4}$$

For fixed $0 < \chi < r$, the polynomial factor is absorbed by $2^{(r-\chi)D_{\mathrm{dens}}m}$, uniformly at cost $O_{r,\chi}(D_{\mathrm{dens}}^{-2})$. Proposition 3.3 absorbs the certification complement after D_{dens} is restricted below a fixed positive threshold. Dyadic summation proves the claimed density update. Iterating it any fixed number of times therefore still gives a natural-density-one set.

Now fix $0 < \alpha < 1$ and $\varepsilon > 0$. Choose $0 < \alpha' < \alpha$ with $c_*(\alpha - \alpha') < \varepsilon/3$, then choose a fixed $R \ge 1$ and $r = (\alpha')^{1/R} > a_0$. Finally choose $0 < \eta < r - a_0$ so that $c_*(1-\alpha')\eta/(1-r) < \varepsilon/3$. On the resulting R-fold pullback set, write $n_0 = n$ and $n_{i+1} = G(n_i)$. Finite startup changes only a constant, so $n_i \le K_* n^{r^i}$ for a fixed K_*, and consequently $n_R \le K_* n^{\alpha'} \le n^\alpha$ eventually. Summing (A.3) gives

$$
\begin{aligned}
k_R &\le \frac{1+\eta-r}{g}\,\frac{1-r^R}{1-r}\log_2 n + O_{\alpha,\varepsilon}(1) \\
&= c_*(1-\alpha')\left(1+\frac{\eta}{1-r}\right)\log n + O_{\alpha,\varepsilon}(1).
\end{aligned} \tag{A.5}
$$

The two strict parameter margins leave a final third of the clock slack to absorb the fixed remainder, proving (1.11). □

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